

Linear Algebra: Matrices, matrix operations, solution of linear equations

Mathematical Models for Data Science

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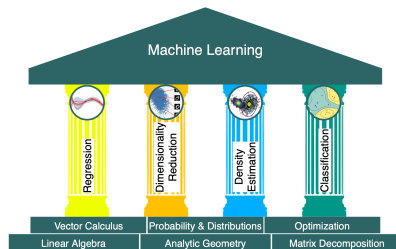
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Motivation I

Decision Making

- Represent of data as vectors
- Choose appropriate model
- Learn from available data to make predictions
 - Critical for decision making

Pillars of Machine Learning



Introduction I

Data versus Vectors

- Not all data is numerical but representation in number format is a useful.
- Assumption: data has been converted into a numerical representation and can be into a computer program.

Vectors representations

- Array of numbers (Computer Science view)
- Quantity with magnitude and direction (Physics view)
- Object that obeys addition and scaling (Mathematical view)

Introduction II

Model

- Process of generating data similar to data set in mind.
- Simplified versions of the real (unknown) data-generating process.
- A good model: Predict accurately what would happen in the real world without performing real-world experiments.

Introduction III

Putting it all together

- Linear Algebra: Study of vectors and matrices.
- Matrix Decomposition -Intuitive interpretation of the data and more efficient learning.
- Optimization: Which direction to look for solutions. Requires calculus.

Unusual vectors I

Examples

- Geometric vectors: These are directed segments.
 - Sum of two geometric vectors is another geometric vector: $\vec{z} = \vec{x} + \vec{y}$
 - Multiplication by a scalar, b is also a geometric vector: $b\vec{x}$
- Polynomials are also vectors: Can be added together or multiplied by a scalar to give another polynomial.
- Audio signals: Represented as a series of numbers. Addition of audio signals gives a new audio signal.
- **Our Focus:** Elements in $\mathbb{R}^n$ (tuples of n real numbers) are also vectors. These also can be added together and multiplied by a scalar.

Systems of linear equations I

Problem

- A company produces products $N_1, \dots, N_n$ for which resources $R_1, \dots, R_m$ are required.
- To produce a unit of product N_j , a_{ij} units of resource R_i are needed, where $i = 1, \dots, m$ and $j = 1, \dots, n$.
- **Objective:** Find an optimal production plan, i.e., a plan of how many units x_j of product N_j should be produced if a total of b_i units of resource R_i are available and (ideally) no resources are left over.

Systems of linear equations II

Approach

For $x_1, \dots, x_n$ units of the corresponding products, the total resource

$$R_i = a_{i1}x_1 + \dots + a_{in}x_n$$

is needed. An optimal production, therefore, satisfies the equations, called system of linear equations with unknowns $x_1, \dots, x_n$ as their solutions

$$a_{11}x_1 + \dots + a_{1n}x_n = b_1$$

$$\vdots$$

$$a_{m1}x_1 + \dots + a_{mn}x_n = b_m$$

Types of solutions I

No Solution

Adding two equations contradicts the third.

$$x_1 + x_2 + x_3 = 3$$

$$x_1 - x_2 + 2x_3 = 2$$

$$2x_1 + 3x_3 = 1$$

Types of solutions II

Unique Solution

Solving the equations to get one solution which satisfy all equations.

$$x_1 + x_2 + x_3 = 3$$

$$x_1 - x_2 + 2x_3 = 2$$

$$x_2 + x_3 = 2$$

Types of solutions III

Infinitely many solutions

More than one solution.

$$x_1 + x_2 + x_3 = 3$$

$$x_1 - x_2 + 2x_3 = 2$$

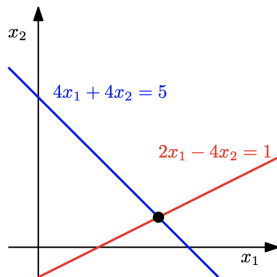
$$2x_1 + 3x_3 = 5$$

Geometrical Interpretation I

Characteristics of solution

- Must satisfy all equations simultaneously, the solution set is the intersection set.
- Intersection set
 - Parallel lines $\implies$ no intersection $\implies$ No solution
 - A point $\implies$ one intersection $\implies$ Unique solution
 - Same line $\implies$ many intersections $\implies$ Infinite solutions

Graphical representation



Geometrical Interpretation II

Compact notation

- Collect the coefficients a_{ij} into vectors

$$\begin{bmatrix} a_{11} \\ \vdots \\ a_{m1} \end{bmatrix} x_1 + \begin{bmatrix} a_{12} \\ \vdots \\ a_{m2} \end{bmatrix} x_2 + \cdots + \begin{bmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{bmatrix} x_n = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

- Collect the vectors into matrices

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

Representation

Definition

- Matrices represent systems of linear equations/functions (linear mappings)
- Matrix $\mathbf{A}$ is an $m \cdot n$ tuple of elements having m rows and n columns, where, $m, n \in \mathbb{N}$. Thus, $\mathbf{A} \in \mathbb{R}^{m \times n}$ represents:

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

- **Convention:** $(1, n)$ -matrices are rows, $(m, 1)$ -matrices are columns.

Matrix Operations I

Addition

The sum of two matrices $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{B} \in \mathbb{R}^{m \times n}$ is defined as the element-wise sum.

$$\mathbf{A} + \mathbf{B} := \begin{bmatrix} a_{11} + b_{11} & \dots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & \dots & a_{2n} + b_{2n} \\ \vdots & \ddots & \vdots \\ a_{m1} + b_{m1} & \dots & a_{mn} + b_{mn} \end{bmatrix} \in \mathbb{R}^{m \times n}$$

Matrix Operations II

Multiplication

- Matrices can only be multiplied if their “inner” dimensions match. An $n \times k$ -matrix A multiplied by $k \times m$ -matrix B gives matrix C as

$$\underbrace{A}_{n \times k} \underbrace{B}_{k \times m} = \underbrace{C}_{n \times m}$$

- Component-wise representation

$$c_{ij} = \sum_{l=1}^n a_{il} b_{lj}$$

- Element-wise multiplication is called **Hadamard product**.

Properties of matrices

Fundamental properties

- Matrix multiplication is not commutative, i.e., $\mathbf{AB} \neq \mathbf{BA}$.
- Associativity: $\forall \mathbf{A} \in \mathbb{R}^{m \times n}, \mathbf{B} \in \mathbb{R}^{n \times p}, \mathbf{C} \in \mathbb{R}^{p \times q}$, i.e., $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$
- Distributivity: $\forall \mathbf{A}, \mathbf{B} \in \mathbb{R}^{m \times n}, \mathbf{C}, \mathbf{D} \in \mathbb{R}^{n \times p}$

$$(\mathbf{A} + \mathbf{B})\mathbf{C} = \mathbf{AC} + \mathbf{BC}$$

$$\mathbf{A}(\mathbf{C} + \mathbf{D}) = \mathbf{AC} + \mathbf{AD}$$

- Multiplication with identity matrix: $\forall \mathbf{A} \in \mathbb{R}^{m \times n}$, i.e., $\mathbf{I}_m \mathbf{A} = \mathbf{A} \mathbf{I}_n = \mathbf{A}$

Inverse I

Definition

- Consider matrix $A \in \mathbb{R}^{n \times n}$. Then $B \in \mathbb{R}^{n \times n}$ is the inverse (A^{-1}) of A if:

$$AB = I_n = BA$$

- If inverse exists, then A is called **non-singular** or **invertible** or **regular** matrix. Otherwise, **singular** or **non-invertible**.

Inverse II

Existence of the Inverse of a 2×2 -matrix

Let $\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ and $\mathbf{A}' = \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$ then,

$$\mathbf{A}\mathbf{A}' = \begin{bmatrix} a_{11}a_{22} - a_{12}a_{21} & 0 \\ 0 & a_{11}a_{22} - a_{12}a_{21} \end{bmatrix} = (a_{11}a_{22} - a_{12}a_{21})\mathbf{I}$$

$$\mathbf{A}^{-1} = \frac{1}{a_{11}a_{22} - a_{12}a_{21}} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$$

if and only if, the determinant of a 2×2 -matrix, $a_{11}a_{22} - a_{12}a_{21} \neq 0$.

Transpose I

Definition

- For $\mathbf{A} \in \mathbb{R}^{m \times n}$ the matrix $\mathbf{B} \in \mathbb{R}^{n \times m}$ with $b_{ji} = a_{ij}$ is the **transpose** of $\mathbf{A}$ denoted by $\mathbf{B} = \mathbf{A}^T$.
- Transpose of $\mathbf{A}$ is obtained by writing columns of $\mathbf{A}$ as the rows of $\mathbf{A}^T$.
- If $\mathbf{A}$ is invertible, then $\mathbf{A}^T$ is also invertible.

Transpose II

Symmetric Matrix

- A matrix $\mathbf{A}$ is symmetric if $\mathbf{A} = \mathbf{A}^T$.
- Only square matrices ($n \times n$) can be symmetric.
- Sum of symmetric matrices is always symmetric.
- Product of symmetric matrix is always defined, it is generally not symmetric.

Properties of inverses and transposes I

Inverses

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{I} = \mathbf{A}^{-1}\mathbf{A}$$

$$(\mathbf{A}\mathbf{B})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$$

$$(\mathbf{A} + \mathbf{B})^{-1} \neq \mathbf{A}^{-1} + \mathbf{B}^{-1}$$

Transposes

$$(\mathbf{A}^T)^T = \mathbf{A}$$

$$(\mathbf{A}\mathbf{B})^T = \mathbf{B}^T\mathbf{A}^T$$

$$(\mathbf{A} + \mathbf{B})^T = \mathbf{A}^T + \mathbf{B}^T$$

$$(\mathbf{A}^{-1})^T = (\mathbf{A}^T)^{-1} = \mathbf{A}^{-T}$$

Scalar multiplication I

Matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ when multiplied by a scalar ($\lambda \in \mathbb{R}$) then, each element of the matrix $\mathbf{A}$ gets “scaled” by λ , i.e., $\lambda \mathbf{A} = \mathbf{K}$ or $K_{ij} = \lambda a_{ij}$. For $\lambda, \psi \in \mathbb{R}$ the following properties hold:

Associativity

$$(\lambda\psi)\mathbf{C} = \lambda(\psi\mathbf{C})$$

$$\lambda(\mathbf{BC}) = (\lambda\mathbf{B})\mathbf{C} = \mathbf{B}(\lambda\mathbf{C}) = (\mathbf{BC})\lambda$$

$$(\lambda\mathbf{C})^T = \mathbf{C}^T \lambda^T = \mathbf{C}^T \lambda = \lambda\mathbf{C}^T$$

Distributivity

$$(\lambda + \psi)\mathbf{C} = \lambda\mathbf{C} + \psi\mathbf{C}$$

$$\lambda(\mathbf{B} + \mathbf{C}) = \lambda\mathbf{B} + \lambda\mathbf{C}$$

Matrix representation I

Example

Consider the equations:

$$2x_1 + 3x_2 + 5x_3 = 1$$

$$4x_1 - 2x_2 - 7x_3 = 8$$

$$9x_1 + 5x_2 - 3x_3 = 2$$

It can be written as:

$$\begin{bmatrix} 2 & 3 & 5 \\ 4 & -2 & -7 \\ 9 & 5 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 8 \\ 2 \end{bmatrix}$$

Matrix representation II

Important features

- System of linear equations can be compactly represented in their matrix form as $A\mathbf{x} = \mathbf{b}$.
- Product, $A\mathbf{x}$, is a linear combination of the columns of A .

Particular solution I

Example

- Consider system of linear equations:
$$\begin{bmatrix} 1 & 0 & 8 & -4 \\ 0 & 1 & 2 & 12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 42 \\ 8 \end{bmatrix}.$$
- If we take 42 times the first column and 8 times the second column, i.e.,
$$b = \begin{bmatrix} 42 \\ 8 \end{bmatrix} = 42 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 8 \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$
- We get **particular** or **special** solution as: $\begin{bmatrix} 42 & 8 & 0 & 0 \end{bmatrix}^T.$

Particular solution II

Important points

- This is not the only solution of this system of linear equations.
- To capture all the other solutions, we need to generate $\mathbf{0}$ in a non-trivial way using the columns of the matrix: Adding $\mathbf{0}$ to the special solution does not change the special solution.
- Expressing the third column using the first two columns as

$$\begin{bmatrix} 8 \\ 2 \end{bmatrix} = 8 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

gives $(x_1, x_2, x_3, x_4) = (8, 2, -1, 0)$

Particular solution III

Scaling the solution

Scaling of $(x_1, x_2, x_3, x_4) = (8, 2, -1, 0)$ by $\lambda_1 \in \mathbb{R}$ produces **0** vector

$$\begin{bmatrix} 1 & 0 & 8 & -4 \\ 0 & 1 & 2 & 12 \end{bmatrix} \left(\lambda_1 \begin{bmatrix} 8 \\ 2 \\ -1 \\ 0 \end{bmatrix} \right) = \lambda_1 (8c_1 + 2c_2 - c_3) = \mathbf{0}$$

Particular solution IV

Scaling fourth column

Expressing the fourth column using the first two columns generates set of non-trivial versions of $\mathbf{0}$, for $\lambda_2 \in \mathbb{R}$, as

$$\begin{bmatrix} 1 & 0 & 8 & -4 \\ 0 & 1 & 2 & 12 \end{bmatrix} \left(\lambda_2 \begin{bmatrix} -4 \\ 12 \\ 0 \\ -1 \end{bmatrix} \right) = \lambda_2 (-4c_1 + 12c_2 - c_4) = \mathbf{0}$$

General solution I

Linear combination of particular solutions

By putting also these together we get the general solutions as :

$$\mathbf{x} = \begin{bmatrix} 42 \\ 8 \\ 0 \\ 0 \end{bmatrix} + \lambda_1 \begin{bmatrix} 8 \\ 2 \\ -1 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} -4 \\ 12 \\ 0 \\ -1 \end{bmatrix} \quad \forall \lambda_1, \lambda_2 \in \mathbb{R}.$$

General solution II

Approach

The general approach consists of the following three steps:

- ① Find a particular solution to $Ax = b$.
 - ② Find all solutions to $Ax = 0$.
 - ③ Combine the solutions from previous steps to get the general solution.
- Neither the general nor the particular solution is unique.

Elementary transformations I

Meaning

Transforming system of linear equations into simpler form by keeping the solution set same.

- Exchange of two equations (rows in the matrix representing the system of equations).
- Multiplication of an equation (row) with a constant λ .
- Addition of two equations (rows).

Elementary transformations II

Consider system of linear equations

$$-2x_1 + 4x_2 - 2x_3 - x_4 + 4x_5 = -3$$

$$4x_1 - 8x_2 + 3x_3 - 3x_4 + x_5 = 2$$

$$x_1 - 2x_2 + x_3 - x_4 + x_5 = 0$$

$$x_1 - 2x_2 - 3x_4 + 4x_5 = a$$

Elementary transformations III

Build augmented matrix

Convert this system of equations into the compact matrix notation $\mathbf{Ax} = \mathbf{b}$ and build the augmented matrix.

$$\left[\begin{array}{ccccc|c} -2 & 4 & -2 & -1 & 4 & -3 \\ 4 & -8 & 3 & -3 & 1 & 2 \\ 1 & -2 & 1 & -1 & 1 & 0 \\ 1 & -2 & 0 & -3 & 4 & a \end{array} \right]$$

Elementary transformations I

Row swapping

$$R_1 \leftrightarrow R_3$$

$$\left[\begin{array}{ccccc|c} 1 & -2 & 1 & -1 & 1 & 0 \\ 4 & -8 & 3 & -3 & 1 & 2 \\ -2 & 4 & -2 & -1 & 4 & -3 \\ 1 & -2 & 0 & -3 & 4 & a \end{array} \right]$$

Addition, multiplication, subtraction

$$R_2 \rightarrow R_2 - 4R_1, \quad R_3 \rightarrow 2R_1 + R_3,$$

$$R_4 \rightarrow R_4 - R_1$$

$$\left[\begin{array}{ccccc|c} 1 & -2 & 1 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 & -3 & 2 \\ 0 & 0 & 0 & -3 & 6 & -3 \\ 0 & 0 & -1 & -2 & 3 & a \end{array} \right]$$

Further transformations I

Subtraction

$$R_4 \rightarrow R_4 - R_2 + R_3$$

$$\left[\begin{array}{ccccc|c} 1 & -2 & 1 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 & -3 & 2 \\ 0 & 0 & 0 & -3 & 6 & -3 \\ 0 & 0 & 0 & 0 & 0 & a+1 \end{array} \right]$$

Subtraction, division

$$R_2 \rightarrow -R_2, R_3 \rightarrow -\frac{1}{3}R_3$$

$$\left[\begin{array}{ccccc|c} 1 & -2 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 & 3 & -2 \\ 0 & 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 & a+1 \end{array} \right]$$

Row-Echelon form I

Augmented matrix is in the row-echelon form

$$x_1 - 2x_2 + x_3 - x_4 + x_5 = 0$$

$$x_3 - x_4 + 3x_5 = -2$$

$$x_4 - 2x_5 = 1$$

$$0 = a + 1$$

Row-Echelon form II

Particular solution

For $a = -1$ this system can be solved. A particular solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ -1 \\ 1 \\ 0 \end{bmatrix}$$

Row-Echelon form III

General solution

- The general solution, the set of all possible solutions is

$$x = \begin{bmatrix} 2 \\ 0 \\ -1 \\ 1 \\ 0 \end{bmatrix} + \lambda_1 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} 2 \\ 0 \\ -1 \\ 2 \\ 1 \end{bmatrix}$$

Reduced row-echelon form I

Definition

A matrix is in reduced row-echelon form if:

- 1 It is in row-echelon form.
- 2 Every pivot is 1.
- 3 The pivot is the only non-zero entry in its column.

Gaussian elimination

An algorithm that performs elementary transformations to bring a system of linear equations into reduced row-echelon form.

The Minus-1 Trick I

Trick in action

- Consider matrix $A \in \mathbb{R}^{3 \times 5}$ which is already in reduced row-echelon form

$$A = \begin{bmatrix} 1 & 3 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 & 9 \\ 0 & 0 & 0 & 1 & -4 \end{bmatrix} \implies \tilde{A} = \begin{bmatrix} 1 & 3 & 0 & 0 & 3 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 9 \\ 0 & 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix}$$

- Augment this matrix to a 5×5 matrix by adding rows of the form at the places where the pivots on the diagonal are missing

The Minus-1 Trick II

General solution

- From this form, read out the solutions of $\mathbf{A}\mathbf{x} = \mathbf{0}$ by taking the columns of $\tilde{\mathbf{A}}$, which contain -1 on the diagonal.

$$\tilde{\mathbf{A}} = \begin{bmatrix} 1 & 3 & 0 & 0 & 3 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 9 \\ 0 & 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix} \implies \mathbf{x} \in \mathbb{R}^5 : \mathbf{x} = \lambda_1 \begin{bmatrix} 3 \\ -1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} 3 \\ 0 \\ 9 \\ -4 \\ -1 \end{bmatrix}, \lambda_1, \lambda_2 \in \mathbb{R}$$

Calculating the inverse I

Objective

To compute the inverse $\mathbf{A}^{-1}$ of $\mathbf{A} \in \mathbb{R}^{n \times n}$, we need to find matrix $\mathbf{X}$ that satisfies $\mathbf{AX} = \mathbf{I}_n$. Then, $\mathbf{X} = \mathbf{A}^{-1}$.

Calculating the inverse II

Approach

- Write the set of simultaneous linear equations of the form, $\mathbf{A}\mathbf{X} = \mathbf{I}_n$.
- Solve for $\mathbf{X} = [\mathbf{x}_1 | \dots | \mathbf{x}_n]$ using augmented matrix notation for a compact representation of this set of systems of linear equations to obtain

$$[\mathbf{A} | \mathbf{I}_n] \rightsquigarrow \dots \rightsquigarrow [\mathbf{I}_n | \mathbf{A}^{-1}]$$

- Bring augmented equation system into reduced row-echelon form. The inverse is obtained on the right-hand side of the equation system.
- Thus, determining the inverse of a matrix is equivalent to solving systems of linear equations.

Inverse using Guassian elimination I

Objective

Find inverse of

$$A = \begin{bmatrix} 1 & 0 & 2 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

Inverse using Gaussian elimination II

Approach

- Write down the augmented matrix:
$$\left[\begin{array}{cccc|cccc} 1 & 0 & 2 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \end{array} \right]$$
- Reduce to row-echelon form by Gaussian elimination. The inverse is obtained on the RHS.

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & -1 & 2 & -2 & 2 \\ 0 & 1 & 0 & 0 & 1 & -1 & 2 & -2 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & -1 \\ 0 & 0 & 0 & 1 & -1 & 0 & -1 & 2 \end{array} \right]$$

Algorithms for solving a SLE I

Using pseudo-inverse

Solving a system of linear equations of the form $\mathbf{A}\mathbf{x} = \mathbf{b}$.

- Determine the inverse $\mathbf{A}^{-1}$, such that the solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$ is given as $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$.
- This is only possible if $\mathbf{A}$ is a square matrix and invertible.
- Else use the transformation:

$$\mathbf{A}\mathbf{x} = \mathbf{b} \iff \mathbf{A}^T \mathbf{A}\mathbf{x} = \mathbf{A}^T \mathbf{b} \iff \mathbf{x} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b}$$

- $(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T$ is called the Moore-Penrose pseudo-inverse.

Algorithms for solving a SLE II

Using iteration

- Let $\mathbf{x}_*$ be a solution of $\mathbf{Ax} = \mathbf{b}$. The key idea of these iterative methods is to set up an iteration of the form

$$\mathbf{x}^{(k+1)} = \mathbf{Cx}^{(k)} + \mathbf{d}$$

for suitable $\mathbf{C}$ and $\mathbf{d}$ that reduces the residual error $\|\mathbf{x}^{(k+1)} - \mathbf{x}_*\|$ in every iteration and converges to $\mathbf{x}_*$. We will introduce norms $\|\cdot\|$, which allow us to compute similarities between vectors.